Rendering & visual enhancement

GRAPHICS WITH MR KELLY

National 5 · CAD & technology · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name the N5 CAD rendering techniques: materials, surface texture and light source
- ☐ explain why rendered images are useful for non-specialist audiences

I DO — Worked example

Read how a full-mark answer is built

A 3D CAD model of a speaker has been made to look realistic.

1 State two techniques used to make the model look realistic. (2 marks)

Full-mark answer each ✓ is one mark

Materials, such as brushed aluminium on the speaker grille ✓. A light source shining from the top left ✓.

Why marks are not awarded

"Colour and shadows."

0 marks: at N5 colour, highlights and shadows are not accepted.

WE DO — Guided practice

Use the starters and word bank to help you

A rendered CAD image of a new watch is used in a magazine advert.

2 Explain two advantages of rendered images for a non-specialist audience. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

materials · non-specialists · colours · textures · realistically · before · understand

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Rendered images are most useful for…
☐ Manufacturers needing sizes ☐ Non-specialist audiences ☐ Hatching sections ☐ Showing hidden detail

b) Which is an accepted N5 CAD rendering technique?
☐ Colour ☐ Highlights ☐ Light source ☐ Shadows

c) A logo applied to a curved surface of a model is a…
☐ Texture ☐ Decal ☐ Material ☐ Light source

4 Key terms: write the letter of the matching description next to each term.

1. **N5 CAD rendering techniques**	____	**A** No. Neither are highlights or shadows.
2. **Is "colour" a rendering technique at N5?**	____	**B** An image such as a logo applied to the surface of a 3D model.
3. **Decal**	____	**C** Materials, surface texture and light source (decals accepted).

A designer 3D prints a prototype of a cookie cutter.

5 Explain two benefits of 3D printing the prototype. (2 marks)

6 Why render for a client? (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?